

Adam Faci

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*PhD in Artificial Intelligence.
Interdisciplinary AI & SSH researcher working on knowledge graphs
and large language models.*

Experience & Education

- Oct. 2023 - **Postdoc in large language models for the SSH,** *CNRS - HNLab, Paris, France.*
Mar. 2027 Interdisciplinary research and research administration within a CNRS *infrastructure de recherche étoile* (IR*).
Leading a use case on an AI research-assistant agent for the social sciences and humanities within the European project LLMs4EU.
- Apr. - Sep. 2023 **Postdoc in ontologies for cybersecurity,** *CNAM, Paris, France.*
Operationalisation of ontologies for incident response & supervision of a PhD student.
- 2019 - 2022 **PhD in Artificial Intelligence,** *LIP6, Sorbonne Université, Paris, France.*
Representation, Simulation and Exploitation of Knowledge in Conceptual Graphs.
- 2015 - 2018 **MSc in Computer Science,** *TELECOM Nancy, Villers-lès-Nancy, France,*
Engineering degree, major in Software Engineering.
Artificial Intelligence, Software Development, Project Management.

Publications

- (Under review) **When Does a Probe Identify a Representation? Semantic Counterfactual Validation in Language Models,** *Faci, A.*
2027 ICLR: International Conference on Learning Representations.
- (Under review) **Query-Type Routing for Retrieval-Augmented Generation: Exact Bounds and Their Dependence on the Evaluation Metric,** *Faci, A.*
2027 ECIR: European Conference on Information Retrieval.
- (Under review) **The Catalogue Is Part of the Model: Archival Categories in Computational Art History,**
2027 *Faci, A. and Breton, A.*
CHR: Computational Humanities Research conference.
- (Under review) **Validating Frozen Representations Under Confounding: Style as Transmission,**
2027 *Faci, A. and Maronet, L.*
WACV: Winter Conference on Applications of Computer Vision.
- (Under review) **SSH-Bench: A French–English Multitask Benchmark for Evaluating Language Models on Scholarly Literature in the Social Sciences and Humanities,** *Faci, A. and Breton, A.*
2027 EACL: Conference of the European Chapter of the Association for Computational Linguistics.
- (Under review) **OntoGraphRag: lunettes ontologiques pour agents conversationnels et dialectique des textes,** *Faci, A. and Provendier, E.*
2027 *Humanités Numériques*, issue 15.
- 2026 **Integrating Knowledge Graphs and Multilingual Scholarly Corpora for Domain-Adaptive LLMs in SSH,** *Faci, A., Frontini, F. and Miaschi, A.*
LLMs4SSH workshop at LREC: Language Resources and Evaluation Conference.
- 2026 **Théorie des genres comparatiste et folksonomies,** *Langlet, I. and Faci, A.*
Littératures et cultures numériques, series Poétiques comparatistes.
- 2026 **Structuration argumentative des débats parlementaires par IA : vers une modélisation interprétable des dynamiques discursives,** *Faci, A.*
Humanistica 2026 proceedings, published in the ACL Anthology.

- 2025 **Lire à différentes échelles. Une analyse multiscalaire de la retrieval-augmented generation en sciences humaines et sociales**, *Silvestre de Sacy, A., Faci, A., Maronet, L. and Pouyllau, S. Humanités Numériques*, volume 12.
- 2025 **Towards a Situational Aware-Based Cybersecurity Remediation**,
Kanawati, R., Lammari, N., Faci, A. and Mimouni, N.
NTMS: IFIP International Conference on New Technologies, Mobility and Security.
- 2021 **Fuzzy Conceptual Graphs: a comparative discussion**, *Faci, A., Lesot, M.-J. and Laudy, C.*
IEEE SSCI-FOCI: Symposium Series on Computational Intelligence, Foundations of Computational Intelligence.
- 2021 **CG2A: Conceptual Graphs Generation Algorithm**, *Faci, A., Lesot, M.-J. and Laudy, C.*
EUSFLAT: Conference of the European Society for Fuzzy Logic and Technology.
- 2021 **cgSpan: Pattern Mining in Conceptual Graphs**, *Faci, A., Lesot, M.-J. and Laudy, C.*
ICAISC: International Conference on Artificial Intelligence and Soft Computing.

Talks

- Sep. 2026 **(Poster) Writing and Evaluating Under an Identity: Gender and Affiliation Bias in Scientific Review Generated by Large Language Models**, *Faci, A. and Breton, A.*
Conference "Modèles et apprentissages en sciences humaines et sociales".
- Sep. 2026 **(Poster) A Review of Fine-Tuning for QA and a Proposal for a New Annotated Corpus**,
Breton, A. and Faci, A.
Conference "Modèles et apprentissages en sciences humaines et sociales".
- Sep. 2026 **Framing the Source: Knowledge Graphs, Generative Models, and the Epistemology of Qualitative Data Exploration**, *Provendier, E. and Faci, A.*
JADH: conference of the Japanese Association for Digital Humanities.
- Sep. 2026 **Each Reading Is a World: Neuro-Symbolic AI Agents as Simulated Readers over a Large Literary Corpus**, *Faci, A. and Schuh, J.*
JADH: conference of the Japanese Association for Digital Humanities.
- Jul. 2026 **Co-Designing an Annotation Infrastructure for Small, Interpretive Datasets with Arkindex**,
Truc, A., Maronet, L., Faci, A., Schuh, J., Charpier, M. and Kermorvant, C.
Digital Humanities conference (ADHO).
- Jul. 2026 **From Distant to Close Viewing: A Frugal Approach for the Study of Ancient Indian Patterns**,
Maronet, L. and Faci, A.
Digital Humanities conference (ADHO).
- Jun. 2026 **Segmentation automatique d'images en histoire de l'art : nouvelles méthodes et réflexion épistémologique**, *Faci, A. and Maronet, L.*
(Invited) VHS international seminar "Étude de la circulation du savoir scientifique par l'image lors des périodes médiévale et moderne".
- May 2026 **(Poster) The Classroom as Methodological Laboratory: Participatory Pedagogy for Digital Humanities Innovation**, *Faci, A.*
DARIAH international conference.
- May 2026 **Structuration argumentative des débats parlementaires par IA : vers une modélisation interprétable des dynamiques discursives**, *Faci, A.*
Humanistica international conference.
- May 2026 **(Session chair) Projets et méthodes pour la récupération et structuration de corpus**,
Faci, A.
Invited to the third Médias 19 congress, "Le journal dans tous ses états".
- Oct. 2025 **Des travaux du HN Lab, d'Arkindex à Nakala : plateformes et outils pour le traitement d'images en SHS**, *Maronet, L. and Faci, A.*
Digit_Hum 2025 workshop, "IA, Intelligences Augmentées pour les SHS", École Normale Supérieure.
- Oct. 2025 **Les marqueurs de genre dans les interactions de la fiction policière (1945-1989)**,
Bouzereau, C., Longrée, D. and Faci, A.
Journées Linguistique de Corpus.

- Jun. 2025 **Talk 1: « Des canons par centaines » : comment les étiquetages manuels et automatiques du web social recomposent-ils les corpus littéraires ?** *Langlet, I. and Faci, A.*
Talk 2: « AI Reading Club » : comment faire lire des agents d'intelligence artificielle « dans le genre », *Bréan, S., Faci, A. and Schuh, J.*
 DH@LLM international conference, Sorbonne U., CRIHN & IEA.
- May 2025 **SegmentArt, une méthode plus rapide pour annoter des images en utilisant SAM2**, *Maronet, L. and Faci, A.*
 Humanistica international conference.
- Nov. 2024 **Les LLMs sont-ils de bons lecteurs ?**, *Schuh, J. and Faci, A.*
 National conference "Un chat à la fac de lettres ?" on the use of large language models for text analysis, Consortium-HN ARIANE.
- Nov. 2024 **Usages raisonnés des IA en SHS**, *Pouyllau, S., Silvestre de Sacy, A. and Faci, A.*
 DHNord national conference on digital humanities.
- May 2024 **(Poster) Feedback et réflexions sur le futur de l'infrastructure de recherche Huma-Num**, *Pouyllau, S., Silvestre de Sacy, A. and Faci, A.*
 DHNB international conference "From experimentation to experience: lessons learned from the intersections between digital humanities and cultural heritage".
- May 2024 **Exploration de corpus fondée sur le RAG**, *Silvestre de Sacy, A. and Faci, A.*
 ColDoc international conference "La linguistique dans une ère nouvelle : discours, méthodes et technologies dans le paysage contemporain".
- May 2024 **Analyse de corpus GLAM via une chaîne RAG**, *Silvestre de Sacy, A. and Faci, A.*
 AI4LAM conference at Stanford (invited), on the use of artificial intelligence in, for and by libraries, archives and museums.
- Apr. 2022 **PhD defence**, *Paris, France.*
 Jean-Gabriel Ganascia, Cyril de Runz, Trevor Martin, Maria Rifqi, Marie-Jeanne Lesot & Claire Laudy.

Teaching

- 2026 *AI research seminar, Master's in Lettres, Savoirs, Création et Numérique.* Univ. Gustave Eiffel
- 2026 *RAG: Contextualised AI. Artificial Interpretations, Alternative Interpretations.* Sorbonne Université
- 2025 *Training: Ethics of LLMs in the Social Sciences and Humanities.* CNRS/CC IN2P3
- 2025 *ANF (CNRS national training school): Working in Laboratories with Generative AI.* Campus Condorcet
- 2024 & 2025 *New Writing and Publishing Devices; AI, Ethics and Journalistic Work.* Université Nanterre
- 2024 *RAG, Annotation and Research Methods in History.* Université d'Évry
- 2024 *LLMs for Research Practices and Ethical Questions.* Huma-Num
- 2018 *Algorithmics & Compilation.* TELECOM Nancy engineering school
- 2013 *Statistics.* Université Panthéon-Sorbonne

Projects

- (In prep.) 2026 **The Geography and Language of Legitimacy: A Controlled Audit of Identity Exposure in LLM Peer Review**, *Faci, A.*
 Special issue "Ethics in NLP and CL" of the journal *Computational Linguistics* (submission deadline: 27 Nov. 2026).
- (In prep.) 2027 **The Model as Gatekeeper: Reviewer and Author Affiliation Bias in LLM**, *Faci, A.*
 ACL: Annual Meeting of the Association for Computational Linguistics.
- Demo **SegmentArt V4**, *with Léa Maronet.*
 Fast annotation aid using SAM2. See <https://github.com/AdamFaci/segmentart>
- European Project **Leading the ReSearch SSH use case of LLMs4EU**, *with Stéphane Pouyllau, Nicolas Larrousse, Francesca Frontini, Alessio Miaschi, Léa Maronet and Pascal Cuxac.*
 Interpretable multilingual large language models as research assistants for the SSH.
- From Invisible Automation to the Conversational Interface: Recompositions of French Media Discourse on Artificial Intelligence since ChatGPT**, *with Émile Provendier.*
 Analysis of a three-year Europresse corpus on generative AI in France, followed by a comparative analysis.
- Modes of Knowledge Representation and Disciplinary Reconfigurations: A Socio-History of the Diffusion of AI in French Digital Humanities (2000–2025).**
 Contrastive study of the rapid diffusion in linguistics and the later diffusion in art history.

Seminar *Interprétations artificielles*,

with Julien Schuh.

Participant in 2025, co-organiser in 2026. Edited volume in preparation.

Ethics in Artificial Intelligence.

Article in preparation on the evaluation of discriminatory biases in generative large language models, proposing AI guidelines for SSH communities.

ANR **Study of Crime Novels (book in preparation),** *with Camille Bouzereau and Sybila Guéneau.*

- Polarisation
2023 - 2025
- Pattern analysis in novels for contrastive corpus description;
 - Violent motifs in novels;
 - Analysis of the structure of novels' opening sentences, talk at Bases Corpus Langage (Université Côte d'Azur) 2024.

Administration **Applications to ANR, Labcom and European calls,** *with Stéphane Pouyllau and Léa Maronet.*
Formalisation of the research strategy and mobilisation of partners.

2024 - 2025 **PhD supervision,** *with Nadira Lammari and Nada Mimouni (CNAM).*
Ontology for cybersecurity incident response; doctoral work of Rayan Kanawati.

Co-directing a PictorIA working group, *with Emmanuel Château-Dutier, Léa Maronet and Alice Truc.*
Image annotation standards in SSH; since November 2024.

2019 - 2021 **Coordination of an inter-doctoral working group,** *THALES, Palaiseau, France.*

Languages

French **Native speaker**

English **Fluent (950 TOEIC)**

PhD Defence Report

Jury: J.-G. Ganascia, C. de Runz, T. Martin, M.-J. Lesot, C. Laudy.

Mr Adam Faci presented his work in a clear and didactic manner, using well-designed slides appropriate to his subject. He then responded to the many questions posed by the jury with relevance, seriousness and competence, without evading any of them. In so doing, he demonstrated his mastery of his research subject.

His thesis is structured around three new contributions aimed at extracting frequent patterns from conceptual graphs and validating this extraction.

The first algorithmic contribution, whose originality the jury wishes to underline, concerns the extraction of frequent patterns from conceptual graphs by means of symbolic information fusion operators and the exploitation of their specificities. The second allows the generation of databases of synthetic conceptual graphs, which the thesis uses to validate the extraction of frequent patterns.

The third provides a rich, critical review of the state of the art on the introduction of fuzzy modalities into conceptual graphs, with a view to extracting frequent patterns from fuzzy conceptual graphs.

Mr Adam Faci brings these three solid contributions together coherently.

For all these reasons, the jury unanimously decides to award the degree of Doctor in Computer Science from the Faculty of Science of Sorbonne University to Mr Adam Faci.